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MILITARY SPECIFICATION

POLYETHER CUSHIONING MATERIAL

FOAM-IN-PLACE, FLEXIBLE

1. SCOPE

1.1 **Scope.** This specification covers a two component liquid polyether system for producing a flexible foam cushioning material for electronic equipment.

1.2 **Classification.** The polyether foam shall be furnished in the following types, as specified (see 6.2).

Types	*Density, lbs/cu ft.*
I	2.0–3.0
II	3.0–4.0
III	4.0–5.0

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 —Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM D 1564 Flexible Urethane Foam

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race St., Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 **Materials.** The materials used in the manufacture of polyether foam shall consist of a polyether resin as a basic ingredient which shall be mixed with a catalyst and cured in accordance with 3.2. The flexible foam cushioning product shall conform to the requirements specified herein.

3.2 **Mixing and curing.** Mixing and curing shall be in accordance with the manufacturer's instructions.

3.3 **Instructions.** Unless otherwise specified, the supplier shall furnish one mixing and curing instruction sheet with each unit package. The instruction sheet shall also contain the following warning:

Caution: **From the time mixing is begun, gases are liberated which are classed as extreme irritants. All mixing, pouring, and foaming must be done in an exhaust hood or areas having adequate ventilation. Any areas of the skin in contact with the mixed**

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or unmixed foam shall be flushed immediately with isopropyl alcohol, or thoroughly washed with soap and water.

3.4 Physical properties. When mixed and cured as specified in 3.2, the material shall be in accordance with table I.

TABLE I

Physical properties	Type I	Type II	Type III
Density, lbs/cu. ft., ± 10%.	2.0-3.0	3.0-4.0	4.0-5.0
Tensile strength, psi, minimum.	20	25	
Elongation, %, minimum	100	130	
Compression set, 22 hrs/70°C.			
ASTM D 1564, Method B %, maximum.	30	30	30
Compression deflection, psi, maximum			
10%	.61	1.28	1.60
25%	.66	1.84	
50%	.96	2.47	

3.5 Identification. The manufacturer shall identify each resin and catalyst container as specified in 5.3.

3.6 Shelf life. Shelf life of the polyether resin and the catalyst shall be not less than 90 days, when stored in their original containers, at a temperature between 65° and 75° F.

3.7 Workmanship. Workmanship shall conform to the best commercial practice for this product. There shall be no defects which might render the material unsuitable for its intended purpose. The compound shall be homogeneous and free from lumps and coarse particles.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Quality conformance inspection shall consist of examination and testing of the polyether cushioning material for conformance to the requirements specified herein.

4.3 Lot. A lot shall consist of polyether resin from the same batch, along with a proportionate supply of the catalyst (see 3.2) from the same batch, and submitted for acceptance at one time.

4.4 Examination. Sampling shall be in accordance with Standard MIL–STD–105, inspection level II, and with an AQL of 2.5 percent defective. Visual examination of the sample containers and contents shall be made for fill, leakage, marking, and workmanship. Preservation, packaging, packing, and marking of section 5 shall be examined under the same sampling plan to determine conformance to the requirements of that section.

4.5 Testing. Physical tests specified in table II shall be performed to assure compliance with the requirements specified in table I. Materials shall be mixed and cured as specified in 3.2 in a wood or metal mold. Clean equipment shall be used in weighing and mixing each pour of foam. Mold surfaces shall be smooth and a release agent applied to facilitate removal of foamed parts.

TABLE II

Physical properties tests	ASTM test method
Density, lbs/cu. ft. ± 10%	D1564
Tensile strength, psi, (min)	D1564
Elongation, % (min)	D1564
Compression set, 22 hrs/70°C.	D1564, Method B, % (max)
Compression deflection, psi (max)	D1564
10%	
25%	
50%	

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4.6 Shelf life. Manufacturers shall submit certification that the polyester resin and the catalyst will meet the requirements of this specification after storage for 90 days in their original containers, at a temperature between 65° and 75° F., signed by a responsible agent of the manufacturer.

5. PREPARATION FOR DELIVERY

5.1 Unit packaging.

5.1.1 *Two-part kits.* Unless otherwise specified in the contract or order, the polyether resin and catalyst shall be packaged in individual containers. The ratio of the quantity contained in the polyether resin container to the quantity contained in the catalyst container shall be the same as the recommended mixing ratio of the polyether resin and catalyst.

5.2 Packing. All exterior shipping containers in the shipment shall contain the same number of the type of unit packages. Shipping containers shall be of uniform size and shall be so designed as to insure that damage is prevented during handling and shipping.

5.3 Marking.

5.3.1 *Base compound containers.* Each polyether resin container shall be durably and legibly marked with the following information.

(a) Title and number of this specification.
(b) Classification (see 1.2).
(c) Manufacturer's name and address.
(d) Manufacturer's product designation.
(e) Manufacturer's batch designation.
(f) Manufacturer's batch designation for the catalyst that the base compound is to be used with.
(g) Stock number
(h) Date of manufacture.
(i) Contract number.
(j) Quantity contained.

5.3.2 *Catalyst container.* Each catalyst container shall be durably and legibly marked with the following information:

(a) Manufacturer's name and address.
(b) Manufacturer's product designation.
(c) Manufacturer's batch designation.
(d) Manufacturer's batch designation for the base compound that the catalyst is to be used with.
(e) Date of manufacture.

5.3.3 *Exterior shipping containers.* Each exterior shipping container shall be durably and legibly marked with adequate instructions to insure that damage during handling and shipping is prevented.

5.3.4 *For military agencies.* In addition to any special marking required in the contract or order, marking of the shipping containers shall be in accordance with Standard MIL–STD–129.

6. NOTES

6.1 Intended use. The polyether cushioning material covered by this specification is intended to be used as a light weight cushioning material for irregular shaped electronic equipment.

6.2 Ordering data. Procurement documents for the material should specify the following:

(a) Title, number, and date of this specification.
(b) Type of polyether foam (see 1.2).
(c) Packaging, if other than specified in 5.1.1.

Custodian:
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Review: GL

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